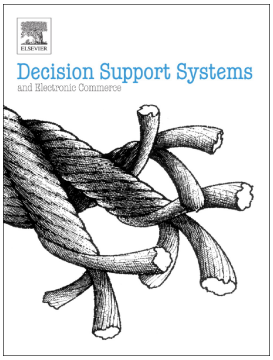
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Predicting Employee Absenteeism for Cost Effective Interventions

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Abstract

This paper describes a decision support system designed for a Belgian Human Resource (HR) and Well-Being Service Provider. Their goal is to improve health and well-being in the workplace, and to this end, the task is to identify groups of employees at risk of sickness absence who can then be targeted with interventions aiming to reduce or prevent absences.

To facilitate deployment, we apply a range of existing machine-learning methods to obtain predictions at monthly intervals using real HR and payroll data that contains no health-related predictors. We model employee absence as a binary classification problem with loss asymmetry and conceptualise a misclassification cost matrix of employee sickness absence. Model performance is evaluated using cost-based metrics, which have intuitive interpretation. We also demonstrate how this problem can be approached when costs are unknown. The proposed flexible evaluation procedure is not restricted to a specific model or domain and can be applied to address other HR analytics questions when deployed.

Keywords: cost-sensitive learning, classification, HR analytics, absenteeism prediction

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1. Introduction

Employee *sickness absence* or *absenteeism*, broadly defined as failure to attend scheduled work as a result of ill health, is a pervasive problem disruptive to operations and costly to the economy. The annual cost of worker absenteeism in the countries of the Organisation for Economic Co-operation and Development (OECD) has been estimated to be between 1.2 and 2% of their total GDP [1], which in current terms translates to between 0.6 and 1 trillion US dollars [2]. It is therefore only natural for employers to seek solutions to this problem.

The motivation for this work was the request of a company in Belgium specialising in

Human Resource (HR) Management and Well-Being for the development of a solution to address employee absenteeism using data science. A direction for finding a possible solution stemmed from the fact that as much as 20% of the working-age population in the OECD countries suffers from common mental illnesses such as anxiety and depression disorders [3], and timely application of preventive measures is crucial in avoiding transition to long-term illness and disability [1]. Interventions in the form of health management or wellness programs have a long history and several meta-analytic studies have reported strong evidence of their effectiveness at reducing employee absenteeism [4–6]. Examples of such interventions can include individual fitness program, stress-management seminars, private or group therapy sessions, and work flexibility arrangements, to name a few.

Naturally, such wellness programs are costly, both in terms of monetary costs for their implementation (which might not always be available due to budget limitation), and hours spent for participation. Therefore, the simplest solution of applying them to all employees at a workplace might not be the most cost-effective. Instead, a decision support system is needed that is able to identify the employees at risk of sickness absence, usually a small fraction of the total workforce, who should then be targeted with a preventive action.

1.1. Related work

HR analytics is most commonly applied to talent acquisition or retention, rather than maintaining and improving the well-being of employees [7]. As observed in [8], the absenteeism prediction literature to date has mostly been concerned with the explanation and association of various risk factors with employee absenteeism, rather than accuracy of predictions. The majority of contributions to the domain of absenteeism prediction come from the field of occupational health and medicine [9–16]. In this field, statistical techniques such as logistic regression and Cox proportional-hazards models are preferred and training data is collected either in the clinical setting or as part of a nation-wide household survey.

Another strand of literature focusses solely on predicting sickness absence using algorithmic models. Amongst those, several studies [17–19] approach employee sickness absence as a regression problem using neural networks that predict hours of absence on the data collected in [19]. [20] uses the same dataset in a multi-class classification setting using decision tree ensembles to predict absences of specific duration.

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The main drawback in all of those studies is lack of transparency with regards to their experimental set-up and model selection procedure. Most contributions apply a single algorithm to a single dataset, the rationale behind the model choice and evaluation being rarely discussed. In contrast, in this paper we consider a wide range of state-of-the-art algorithms and relevant evaluation measures.

Previous studies involved data collection in several waves, usually annually. However, predictions with a horizon of one year are of little relevance to businesses that are trying to reduce direct and indirect costs incurred through loss of productivity and disruption to operations. What is more desirable is the ability of the management to obtain reliable predictions at operationally practical intervals (such as a month or a quarter) preferably with data that is readily available from HR and payroll records. An attempt has been made in [21] to predict sickness absence using only such data from one industry sector, albeit at the impractical one-year prediction horizon. The data we have been provided with for this work does not contain health-related or attitudinal information either, but includes several sectors and allows us to consider a more practical prediction horizon of one month.

Finally, as already mentioned, absenteeism is an event of relative rarity with only a small fraction of the population falling out of the workforce in any given period. Most absenteeism prediction papers do demonstrate an imbalanced class distribution of the outcome variable.

Not accounting for this class imbalance in predictive models has immediate negative implications for model performance [22, 23]. To the best of our knowledge, [24] is the only paper that recognises the importance of class imbalance correction and models sickness absence as a problem with loss asymmetry where cost of misclassifying an absentee is set to the number of non-absentees in the dataset, and vice-versa. This heuristic is commonly applied to treat class imbalance alone,but could be suboptimal when real costs can actually be specified. Specifying them is what we do in an attempt to simultaneously address class imbalance and cost asymmetry using cost-sensitive learning.

1.2. Our contribution

In this work we investigate the use of predictive analytics as a decision support system for increasing employee well-being in the workplace by identifying groups of employees at risk of sickness absence that should be targeted with a wellness intervention.

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Firstly, we employ cost-sensitive learning to treat the unequal misclassification costs pertaining absenteeism, something we could not find in the existing literature. Towards enabling this, the first step and our main contribution is the conceptualisation of a relevant misclassification cost matrix, which can be generalised to a variety of institutional and legislative contexts, and consequently to other datasets as well. A core element of our cost matrix is information on the effectiveness an intervention has on individuals, which is currently lacking and therefore identified as an important direction for future work that requires further collaboration between academia and industry. We also develop business-friendly cost-based evaluation metrics that have an intuitive interpretation.

Secondly, since our data did not contain information on one of the parameters of our cost matrix, we also considered a cost-insensitive approach based on balanced accuracy, where we assume that misclassification errors have equal costs overall, despite the class imbalance.

Finally, we try to illustrate best practices in the domain of absenteeism prediction for the practitioner. For this, we use an anonymised dataset that a Belgian HR and Well-Being company provided us, which contains employee payroll information only without any health-related data. We specify different cost-matrices by considering different realistic scenarios for interventions and their (fictional) effectiveness, and consider a practical prediction horizon of one month. We follow a rigorous experimental design to prevent over-fitting and develop a flexible model-agnostic evaluation framework. In the end, the cost effectiveness of both cost-sensitive and cost-insensitive models is compared on the same cost-matrices to provide an indication on the advantage any of the two approaches might have over one another.

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2. Preliminaries

In what follows we will briefly outline the challenges of classification on imbalanced data. Readers familiar with the material are invited to continue to section 3.

A classifier is a function $f : \mathbf{x} \rightarrow \hat{y}$ that maps a vector of real-valued predictors $\mathbf{x} \in \mathbb{R}^n$ to a predefined target class $\hat{y} \in \mathbb{R}$ based on a training set of data with known true class labels $y \in \mathbb{R}$. In a binary classification setting, the class labels are commonly denoted as $y, \hat{y} \in \{0,1\}$, with the rare class referred to as "positive" and labelled as 1. The typical loss function adopted to measure a classifier's prediction accuracy is the 0-1 loss function, which counts the instances of incorrect

classification.

More detailed error analysis can be conducted using a confusion matrix, an example of which is shown in Table 1. Each entry in the confusion matrix represents the number of observations in the test set that were classified either correctly or incorrectly. The error count is thus split between two error types: false positives and false negatives.

Table 1: Confusion matrix. Each entry represents the number of observations in the respective category on a given test set.

	Predicted 1	**Predicted 0**
True 1	True Positive (TP)	False Negative (FN)
True 0	False Positive (FP)	True Negative (TN)

In most domains different misclassification errors entail different costs, and therefore the question of which of the error types is more costly is determined by the area of application. For example, in the domain of email spam identification, the cost of marking a legitimate email as spam (a false positive) is higher than the cost of missing a spam message (a false negative) [25, 26]. In contrast to that, in the domain of medical diagnostics misclassifying a sick person as healthy (a false negative) and subsequently failing to treat them could entail serious consequences for the individual under consideration (as well as the healthcare system) and is therefore considered more costly than a false positive.

When the problem under consideration suggests cost asymmetry (different error types have different misclassification costs) and the class distribution is skewed, a cost-sensitive classification approach becomes appropriate [22]. Cost-sensitive learning translates the error-minimisation problem to cost-minimisation, where each prediction type (as defined in the confusion matrix) is assigned a cost by means of a misclassification cost matrix $\begin{pmatrix} C_{TP} & C_{FN} \\ C_{FP} & C_{TN} \end{pmatrix}$.

Misclassification costs can be either class-dependent, where all observations of a class entail identical costs, or they could be record-dependent in which case every observation i has its own cost matrix $\begin{pmatrix} C^i_{TP} & C^i_{FN} \\ C^i_{FP} & C^i_{TN} \end{pmatrix}$, derived from a given dataset. As was noted by Weiss [22], appropriately weighing the false negatives should improve classifier performance with respect to

the rare positive class.

2.1. Cost-sensitive decision-making

Most binary classifiers produce predictions in two stages: First a $score \in [0,1]$ is produced for each observation. Then, an instance i is classified as positive ($\hat{y}_i = 1$) if its score is greater than a threshold T, and as negative otherwise. Cost-insensitive classifiers implicitly assume $T = 0.5$ [27], which typically results in poor classifier performance under class imbalance [22]. This can be solved by applying a meta-learning approach called *thresholding* [28], where instead of specifying one decision threshold, a search is performed across all scores produced by the classifier. This approach calculates loss of choice on all scores produced by a classifier. The score at which loss is minimised becomes the decision threshold. The advantage of *thresholding* is that it requires no modification to the base algorithm and can be easily implemented in practice.

An important result derived by Elkan [29] is that in the case where misclassification costs are known, any classifier can be made cost-sensitive by adopting a decision threshold that incorporates these costs, a method referred to as *Direct Minimum Expected Cost Classification (DMECC)* [30, 31]. Here, the decision threshold is defined as follows: $T_{cs} = \frac{C_{FP} - C_{TN}}{C_{FP} - C_{TN} + C_{FN} - C_{TP}}$, where cs stands for cost-sensitive and the threshold can be either class- or record-dependent. This expression can be simplified if we transform the cost matrix into

$$\begin{pmatrix} C'_{TP} = 0 & C'_{FN} = C_{FN} - C_{TP} \\ C'_{FP} = C_{FP} - C_{TN} & C'_{TN} = 0 \end{pmatrix}, \text{ and hence } T_{cs} = \frac{C'_{FP}}{C'_{FP} + C'_{FN}}.$$

One of the main challenges of adopting this approach is defining misclassification costs, which are usually problem- and domain-specific.

2.2. Classifier performance evaluation under class imbalance

As was noted in [22], the use of improper evaluation metrics is one of the pitfalls of dealing with class imbalance, concerning both the assessment of a model's generalisation performance and model selection.

Model generalisation is commonly assessed using one or several accuracy-based metrics, which do not take potential differences in error costs into account. This makes them ill suited for evaluating performance under cost asymmetry.

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One tool that does combine per-class performance is the receiver operating characteristic curve (ROC), which plots true positive rate against false positive rate across all empirical probability estimates produced by a given classifier. The area under the ROC curve (AUC) is commonly used to both assess the generalisation performance of a given classifier and for model selection. ROC graphs are insensitive to changes in class distribution [32, 33], but they do not make the distinction between differences in misclassification error costs [34, 35], making them inappropriate under cost asymmetry. While AUC is useful as a measure that summarises per-class performance across all empirical thresholds, it does not specify which of these thresholds is optimal. To address this shortcoming, one could locate the optimal point on the ROC curve using the balanced accuracy score [36], defined as the arithmetic mean of class-specific accuracy measures: $BACC = \frac{1}{2}(\frac{TP}{P} + \frac{TN}{N})$, where P and N refer to the number of positives and negatives on a given test set.

It has been shown that combining this metric with meta-learning approaches such as *thresholding* can result in performance improvement on imbalanced datasets [37].

In the cost-sensitive literature classifier performance is typically measured in terms of the total expected misclassification costs [29, 38]. Let S_{FP} be the set of false positives produced by a given classifier on a given test set, and let S_{FN} be the set of false negatives. The total misclassification cost of a given classifier is $TC = \sum_{i \in S_{FP}} C^i_{FP} + \sum_{i \in S_{FN}} C^i_{FN}$.

In this paper we will attempt to identify the best practice in the domain of absenteeism prediction, by comparing model selection using error-based versus cost-based metrics.

3. A cost matrix for employee absenteeism and well-being interventions

In this section we present our conceptualisation of a cost matrix of the direct costs of employee sickness absence in relation to a well-being intervention. We consider this one of the main contributions of this paper.

In any period M, an employee is contractually obligated to supply t_M hours of work in return for remuneration W, yielding the *base hourly rate* $\frac{W}{t_M}$ of the employee for this period. If in

this period the employee is absent due to sickness for a total duration of $t_s \in [0, t_M]$ hours, the number of worked hours is reduced to $t_M - t_s$, which are remunerated as usual according to the base rate. However, depending on the legislation of the country of employment, the employer may also be required to remunerate the t_s hours of sickness according to a proportion $r \in [0,1]$ of the employee's base rate[1]. The hourly rate is in this case equal to $\frac{(t_M - t_s)\frac{W}{t_M} + t_s r \frac{W}{t_M}}{t_M - t_s} = (1 + \frac{t_s r}{t_M - t_s})\frac{W}{t_M}$, which is higher than the base rate by a factor of $\frac{t_s r}{t_M - t_s}$, reflecting the loss of productivity associated with the employee's absence.

Suppose now that the employer decides to put the employee through a well-being intervention in an attempt to prevent potential sickness absence. The price of the employee's participation to such an intervention is C, which burdens the employer. In addition, if the intervention requires attendance (such as a coaching seminar) of duration t_i, the number of hours worked by the employee is reduced by as much. Therefore, the hourly rate of an employee who was not going to be absent but is put through an intervention is $\frac{W + C}{t_M - t_i}$, which is also higher than the base rate. If, however, the employee was going to be absent, in addition we expect that the intervention would have a positive effect and result in the reduction of the absence period by $\tilde{t}_s \in [0, t_s]$ hours[2]. The resulting hourly rate is then equal to $\frac{(t_M - t_s + \tilde{t}_s)\frac{W}{t_M} + (t_s - \tilde{t}_s) r \frac{W}{t_M} + C}{t_M - t_s - t_i + \tilde{t}_s} = \frac{W + C}{t_M - t_s - t_i + \tilde{t}_s} - \frac{(t_s - \tilde{t}_s)(1 - r)}{t_M - t_s - t_i + \tilde{t}_s}\frac{W}{t_M}$. Table 2 summarises these hourly rates.

Remark 1. *Clearly, a necessary condition for the intervention to be cost-effective for the employer is that* $t_i < \tilde{t}_s$.

Remark 2. *A limiting factor in specifying a concrete cost matrix is the parameter* $\tilde{t}_s$*, which is a priori unknown and no indication of example values can be found in the literature.*

[1] In some countries, r may actually be non-constant but instead a function of t_s. Our arguments can be extended to such case as well, but for the sake of clarity here we focus on constant r.

[2] For simplicity we assume that $\tilde{t}_s$ can not be observed beyond the prediction horizon of one period ahead, and we leave the dynamic case to future work.

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Table 2: Hourly rates of employee remuneration with respect to a well-being intervention. W represents the employee's salary, t_M the expected work hours, t_s is the absence duration in hours, r is the fraction of the base rate $\frac{W}{t_M}$ to which hours of absence are remunerated as guaranteed by law in the form of statutory sick pay, $\tilde{t}_s$ is the reduction in hours of sickness absence because of the intervention, C is the cost of the intervention, and t_i is the duration of attendance in hours associated with the intervention.

	Intervention	**No Intervention**
Absentee	$C_{TP} = \frac{W+C}{t_M - t_s - t_i + \tilde{t}_s} - \frac{(t_s - \tilde{t}_s)(1-r)}{t_M - t_s - t_i + \tilde{t}_s} \frac{W}{t_M}$	$C_{FN} = (1 + \frac{t_s r}{t_M - t_s}) \frac{W}{t_M}$
Non-Absentee	$C_{FP} = \frac{W+C}{t_M - t_i}$	$C_{TN} = \frac{W}{t_M}$

3.1. The case of Belgium

As we mentioned in the introduction, this work was motivated by a Belgian HR and Well-Being Specialist. We therefore adapt Table 2 to the specifics of Belgian legislation.

In Belgium, throughout all sickness absences lasting up to 30 calendar days, white-collar workers receive full wage equivalent sickness benefits from the employer. As soon as the duration of absence is longer than one calendar month, the benefits are paid by the social security instead. Blue-collar workers receive reduced compensation starting from week two of absence: the employer continues to cover some fraction r of the full wage W, while the remainder is covered by social security[39].

Our sample contains only white-collar employees. By consequence, the parameter $r = 1$ and t_s is defined as the sum hours of sickness absences covered by the employer in any given month M, with the necessary condition that $t_s < t_M$.

Using the hourly rates form Table 2, and after applying the transformation as discussed in section 2.1, such that the cost of correctly classifying observations is zero, we obtain the following costs:

$$\begin{pmatrix} C'_{TP} = 0 & C'_{FN} = \frac{W}{t_M - t_s} - \frac{W+C}{t_M - t_s + \tilde{t}_s - t_i} \\ C'_{FP} = \frac{W+C}{t_M - t_i} - \frac{W}{t_M} & C'_{TN} = 0 \end{pmatrix} \quad (1)$$

To simplify notation we omit the usage of superscripts i, that indicate that costs are record-dependent.

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Remark 3. *Note that while* C'_{FP} *is always positive,* C'_{FN} *may become negative when* $t_i > t_s \geq \tilde{t}_s$. *This violates the so called 'reasonableness' condition defined by Elkan [29], which states that the cost of correct predictions should always be less than the cost of misclassifying, otherwise it is more profitable to misclassify than to classify correctly. A negative misclassification cost is a benefit for the employer, meaning that it is more cost-effective* ***not*** *to apply an intervention to that individual (e.g. due to low expected sickness absence hours).*

4. Experimental framework

In this section we present the experimental procedure that was used to conduct our analysis. Our experiments consist of two parts: in the first instance we predict employee sickness absence using cost-sensitive learning and the cost matrix defined in equation 1, we evaluate model performance using custom cost-based metrics. Then, in view of lack of data regarding the $\tilde{t}_s$ parameter, we additionally predict using cost-insensitive models, which we evaluate using the balanced accuracy score. In both cases, we also report standard metrics such as the AUC.

4.1. Data

Our data contains HR and payroll records from roughly 280 small, medium and large Belgian firms from a variety of industry sectors. The data spans the period between January 2018 and March 2019. We adopt the prospective study design so commonly found in absenteeism prediction literature, where attributes from period M_t are used to predict the outcome in period M_{t+1}. For the sake of reproducibility, we provide details of our data in Table 3.

Table 3: The data we have available, where attributes are from month M_t and targets are from month M_{t+1}

	Attributes	Target	# employees (from # firms)	# positives (%)
1	2018-01	2018-02	50729 (284)	8193 (16.15)
2	2018-02	2018-03	49459 (281)	8161 (16.50)
3	2018-03	2018-04	48202 (280)	4268 (8.85)

4	2018-04	2018-05	51998 (280)	4682 (9.00)
5	2018-05	2018-06	51483 (280)	5123 (9.95)
6	2018-06	2018-07	50843 (281)	3845 (7.56)
7	2018-07	2018-08	51372 (282)	3863 (7.52)
8	2018-08	2018-09	51738 (282)	5201 (10.05)
9	2018-09	2018-10	51130 (282)	6340 (12.40)
10	2018-10	2018-11	50744 (281)	5905 (11.64)
11	2018-11	2018-12	49659 (268)	4986 (10.04)
12	2018-12	2019-01	47128 (268)	6256 (13.27)
13	2019-01	2019-02	52637 (268)	8432 (16.02)
14	2019-02	2019-03	51751 (268)	6493 (12.55)

4.1.1. Target variable

In any given month, each employee's hours of sickness absences were summed and converted to a binary target variable: observations with zero hours of sickness are coded as 0 and 1 otherwise. The resulting distribution of our target variable is highly imbalanced, ranging between 7.5% and 16.5% of positives in any given period.

4.1.2. Predictors

Our dataset consists of 66 features. These include three main groups: socio-demographic controls, work environment controls and employee absence history. One of the novelties of our work is to exploit the rich absence pattern data at our disposal. The main difficulty we are faced with is the absence of any health-related predictors in our dataset.

Demographic features: Employee's demographic features, such as age, gender, marital status, education etc.

Work environment features: Features that describe various aspects of the work circumstances of employees (*wage, contract type, etc.*). We also include fatigue inducing factors such as work shift irregularities (e.g. *weekend work, overtime, night shifts*) as well as patterns of holiday applications (e.g. *holiday frequency and duration, number of rejected holiday applications, time since last holiday of a certain duration*).

Historic absence patterns: measures of recency (time since last absence) and frequency of illnesses, average hours of sickness absences in the 12 months prior to the prediction period and

since the start of employment contract.

4.1.3. Data preparation

Numerical predictors had been cleaned, where appropriate: all values exceeding plausible minima and maxima in a given observation period had been removed (such as age values below 18 and above 100). All levels in categorical predictors had been transformed to binary variables and missing values had subsequently been replaced with zeroes. In continuous numeric variables missing values have been imputed with sample median. In recency variables missing values had been replaced with 366 indicating that the last incident was registered more than one calendar year ago.

4.2. Methods

4.2.1. Classification algorithms

We adopt a wide range of decision tree ensembles in our framework and combine them with state-of-the-art solutions to the problem of class imbalance. Since a requirement for our decision support system was that it could be readily implemented in the cloud, we apply methods with existing open-source implementations

The base algorithm adopted in our experiments is a decision tree classifier (specifically CART [40]). Decision trees have been shown to have a number of highly desirable properties: they can handle mixed data types and missing values; they are insensitive to monotone transformations of the feature space and do not require normalisation of predictors; they can handle irrelevant predictors and are robust to outliers [41]. These properties combined with model interpretability make decision trees highly suitable for business applications. While a single decision tree may not show the highest performance because of high variance, combining a collection of trees into an ensemble decreases variance and improves performance, though at the expense of reduced interpretability [42].

All of the algorithms applied in our experiments are cost-insensitive by default, but can be made cost-sensitive in the presence of explicitly defined costs. In this work we limit ourselves to pre-training and post-training cost-sensitive learning methods as categorised in [43], Table 4 provides a summary.

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Table 4: Overview of the methods used in our analysis. We use the combination of each of the algorithms with one of the sampling and/or calibration methods. CS-SampleEnsemble and CS-SampleBoost methods, naturally, are only combined with calibration. In addition to classifying at the default threshold, we optionally apply *DMECC* to cost-sensitive models and *thresholding* to cost-insensitive models for classification decision. Note: using the name 'Balanced Random Forest' in both cost-sensitive and cost-insensitive models we abuse the terminology slightly, in fact cost-sensitive models perform sampling according to the cost matrix.

Our notation	Method	Parameter settings
Algorithms		
dt	Decision Tree (CART [40])	
Ensembles		
bag	Bagging [44]	#trees = 100
rdf	Random Decision Forest [45]	#trees = 100
rf	Random Forest [46]	#trees = 100
xrf	Extremely Randomised Trees [47]	#trees = 100
adab1	AdaBoost [48]	#nodes=1, #trees = 50
adab2		#nodes=2, #trees = 50
CS-WeightedEnsemble		
wrf	Weighted Random Forest [49]	#trees = 100
CS-SampleEnsemble		
bal_rf	Balanced Random Forest [49]	#trees = 100
bal_rdf	Balanced Random Decision Forest	#trees = 100
easy_ensmb	Easy Ensemble [50]	#estimators = 50
CS-SampleBoost Methods		
rusboost	RUSBoost [51]	#trees = 50
Sampling Methods		
ros	Random Oversampling	
rus	Random Undersampling	
smote	SMOTE [52]	k=5
Calibration		
isotonic	Isotonic Regression [53]	
sigmoid	Platt [54]	
Post-training methods		

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$T = Tbest$	Thresholding [28]	
$T = T_{cs}^{i}$	DMECC [31]	

Class imbalance correction

Each of the cost-insensitive decision tree ensembles can be combined with a sampling method in order to compensate for unequal class distribution of the training data. This can be achieved either by modifying the training set using cost-sensitive sampling and passing it to a classifier of choice (also referred to as *CS-preSampleEnsemble*). Alternatively sampling can be performed at the level of the ensemble (also referred to as *CS-SampleEnsemble*), where both classes are sampled in appropriate proportions prior to training each of the base classifiers in the ensemble.

We performed cost-sensitive data sampling using average training misclassification costs, defined as follows: $\overline{C}_{FP} = \frac{1}{N}\sum_{i \in S_N} C_{FP}^{i}, \overline{C}_{FN} = \frac{1}{P}\sum_{i \in S_P} C_{FN}$. Here S_N is the set of negatives (of size N) and S_P is the set of positives (of size P). The cost ratio $\frac{\overline{C}_{FN}}{\overline{C}_{FP}}$ is used to scale the number of positives for over-sampling, $\frac{\overline{C}_{FP}}{\overline{C}_{FN}}$ is used to scale the number of negatives for under-sampling. [43].

Where costs are unknown, it is reasonable to assume that the optimal class distribution is uniform and the classes are sampled in equal proportions [29].

Calibration

As was observed in [55] and [31], decision tree methods do not produce reliable posterior class membership probability estimates. This has negative implications for application of post-training methods such as described above, where class membership probabilities are used for decision-making[53]. In this work, we optionally apply two probability calibration methods to each of the classifiers in our framework: isotonic regression [53] and Platt scaling [54]

Post-training methods

When costs are record-dependent and are known at the time of estimation *DMECC* can be applied to derive the total misclassification cost on a given test set. If costs are unknown, *thresholding* can be applied to optimise over an error-based loss, which assumes that costs are class-dependent.

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All of our experiments consider the default threshold $T = 0.5$ along with the post-training methods that are simple to implement in practice.

Cost-sensitive models are evaluated at the decision threshold obtained using the *DMECC* approach (explained in section 2.1). For the application of *DMECC*, both types of misclassification costs from equation 1 need to be specified to calculate record-dependent decision thresholds T_{cs}^{i}. Since t_s is unknown at the time of prediction, we estimate this in two ways. First, we set the t_s equal to each record's individual average sickness duration in 12 months prior to prediction (denoting the resulting model $T_{cs}^{i} - mean$). Second, we predict t_s using a Random Forest regressor, trained on positive records from the same training set the cost-sensitive classifier is trained on (denoted $T_{cs}^{i} - postreg$). Cost-insensitive models classify using the optimal decision threshold $T = T_{best}$ obtained using *thresholding* (see section 2).

4.3. Software used

All experiments were conducted using Python (version 3.6.6), the majority of implementations come from *scikit-learn* (version 0.20.2) library [56], implementations of all pre-training methods come from *imbalanced-learn* library [57], post-training methods are our own implementations; the results were processed using Python and *csvkit* [58].

4.4. Experimental set up

4.4.1. Data partition

Three datasets, that contain the same set of employees, served as input to the experimental framework. We randomly split the set of employees into three disjoint sets. The first contained 60 % of all observations and was used for training. The second contained 20 % of the total, used for optimal threshold search respectively, where features were from period M_t and labels were from M_{t+1}. The remaining 20 % of employees were used for testing with features from M_{t+1} and labels from M_{t+2}. This process was repeated 50 times using different random seeds to generate data splits and the results were averaged across iterations.

4.4.2. Training costs

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The definition of the cost parameters are as introduced in section 3.

The values for W, t_M and t_s are readily available from our data. In the absence of information regarding the effectiveness of chosen interventions in reduction of sickness absence, we assume some arbitrary values for the parameter $\tilde{t}_s$. We consider three alternatives for this parameter: a) $\tilde{t}_s = t_s$, b) $\tilde{t}_s = 0.5 \cdot t_s$, and c) $\tilde{t}_s^i = p^i \cdot t_s^i$ where $p^i \in [0,1]$ is an individual percentage of reduction in sickness hours, randomly drawn from a uniform distribution. This parameter was drawn once and was reused across all intervention scenarios. We expect that alternative c) reflects the reality best, but further experiments are needed (see section 6.1).

We consider three examples of interventions with corresponding price C and duration t_i :

Case 1: Fit Check-Up. Aimed at increasing physical activity levels of employees. A professional coach examines one's fitness level using specialised equipment and designs a personalised 12-week fitness programme. The parameter settings are as follows: $C = 100$ (EUR), $t_i = 4$ (hours).

Case 2: A sleep and fitness tracking device. Aimed at promoting healthy lifestyle choices. The built-in software informs on sleep and activity patterns and encourages participants towards positive change in behaviour. The parameter settings are as follows: $C = 30$ (EUR), $t_i = 0$ (hours).

Case 3: Psychotherapy. The employer offers financial support for individual psychotherapy sessions to help reduce stress and prevent burn-out. The parameter settings are as follows: $C = 60$ (EUR), $t_i = 1$ (hour).

4.4.3. Evaluation

Cost-sensitive model performance is typically assessed using the model's total misclassification cost (TC), which sums record-dependent costs on a given test set. In the absence of a decision support system, the employer has two naive solutions, namely not applying any intervention, and applying the intervention indiscriminately to all employees, with respective costs TC_{none} and TC_{all}.

An intervention campaign proposed by a predictive model is worthwhile only if it improves on either of these naive costs, something we are trying to capture via the *Cost*

Improvement Score: $CIS = 1 - \frac{TC}{min(TC_{none}, TC_{all})}$, with the following interpretation. $CIS < 0$ indicates that the campaign predicted by the model is more costly than either of the naive approaches A perfect model that correctly classifies everyone achieves $CIS = 1$. To evaluate whether or not the size of the intervention campaign prescribed by the model is cost-effective, as well as to compare models with similar cost performance, we use *Return On Investment* (ROI). In the context of absence prediction, ROI is defined as the intervention profit (benefits minus costs) over intervention costs:

$$ROI = \sum_{j \in S_{PP}} \frac{\tilde{t}_s^j \frac{w^j}{t_M^j}}{C + t_i \frac{w^j}{t_M^j}} - 1$$, where S_{PP} is the set of positives predicted by the model and j refers to the fact that both the costs and the profits are record dependent. $ROI < 0$ indicates that the intervention campaign prescribed by the model is not cost-effective. Both metrics should be considered in the final model selection. For example, if one model gives CIS 0.7 and ROI is some positive value A, and another model gives CIS is 0.68 with ROI $B > A$, then the second model might be preferred as it requires less budget to be available for similar cost performance.

5. Results and discussion

The primary interest of our work lies in applying cost-sensitive learning to the problem of absenteeism prediction on real data, using our newly designed cost matrix. We evaluate model performance per period, based on the two cost metrics: CIS and ROI (defined in section 4.4.3). Related to this are two further questions: first, whether the usage of record-dependent costs offers any advantage over class-dependent costs, and second, whether predicting the value of t_s using a regressor offers any advantage over using mean historic sickness per individual. We also discuss which of the interventions considered offers the lowest misclassification cost. With the given current difficulty in obtaining all the necessary parameters of the cost matrix, we also try to see whether selecting models using a cost-insensitive metric would be a good alternative to cost-sensitive learning.

Analysing the performance of specific algorithms lies outside the scope of this paper.

5.1. Cost-sensitive absenteeism prediction

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The first part of our experiments provides an illustration of how our proposed misclassification cost matrix of employee absenteeism (defined in equation 1) performs on real world data.

5.1.1. Model performance and cost-effectiveness

Our results show that in every prediction period and intervention combination, we find cost-sensitive models with $CIS > 0$, and with $ROI > 0$.

Figure 1 demonstrates cost performance of the top-ranking models (when ranked by CIS). We note that the largest cost improvement (the difference between the benchmark and the total cost) of the model was achieved under Case 1 and Case 3, while Case 2 - the least expensive intervention - shows only marginal improvement. We also found that $DMECC$ models that used regression predictions to classify, achieved higher cost improvement much more frequently than models that used average hours.

Record- versus class-based costs

As was mentioned in remark 3, in our cost matrix, the reasonableness condition only holds for those individuals j, whose $\tilde{t}_s^{\,j} > t_i$, i.e. when the expected reduction in hours of absence due to the intervention exceeds the duration of the intervention. $DMECC$ approach can directly account for this condition, which allows us to avoid targeting individuals for whom an intervention does not pay off. Models that use a constant threshold for classification decision, do not make such a distinction, and are less frequently ranked among the top models.

Another drawback of using class-based costs became apparent when we applied sampling methods in combination with averaged record-dependent costs (explained in section 4.2.1). We observed that under certain operating conditions, for a number of employees in our sample holds that $C_{FP}^i > C_{FN}^i$, and when the proportion of such individuals in the training sample was large enough, this resulted in: $\overline{C}_{FP} > \overline{C}_{FN}$. Under such cost imbalance, an oversampler will oversample the more costly class i.e. the negatives and an undersampler will undersample the less costly class i.e. the positives, which would increase class imbalance, instead of correcting it. Such models failed in the pre-training phase and were not considered in the rankings.

5.1.2. Cost-effectiveness of interventions

Another question a practitioner might be interested in is: which intervention offers the highest

savings in any given period? To determine this, we shortlist the intervention that yielded the highest CIS on any given prediction period and $\tilde{t}_s$ combination. These results are presented in Table 5. We note that the highest CIS is not necessarily associated with a positive ROI, something we correct by finding a model that has both $ROI > 0$ and $CIS > 0$.

Table 5: Selecting the most cost-effective intervention. For the final decision of which of the interventions offers the highest cost improvement in any given prediction period, we have selected models with the highest CIS and positive ROI. All values are averages over 50 iterations. Where two entries exist in the same period and $\tilde{t}_s$ combination, it means that the model with the highest CIS had $ROI < 0$ and an alternative model was selected instead. The column titled "% pos with $C_{FN} < 0$" shows the percentage of individuals in the test set whose misclassification costs violate the reasonableness conditions (see remark 2)

Target	Intervention		Method	%pos	%pos with $C'_{FN} < 0$	Error-rates		CIS	Cost-based metrics		ROI
period	$\tilde{t}_s$	Case	algo_sampling_calib-T			FNR	FPR		TC	Benchmark	
2018/03	1	1	xrf_rus_sigmoid-T_{cs}^{i} -postreg	15.62	7.94	0.34	0.46	0.21	12756.80	16164.63	0.21
2018/03	0.5	1	easy_ensmb_none_none-$T = 0.5$	15.62	26.59	0.59	0.19	0.18	9545.19	11695.74	0.41
2018/03	rand	1	dt_ros_isotonic-T_{cs}^{i} -postreg	15.62	40.61	0.81	0.13	0.98	384.20	16164.63	1.01
2018/04	1	1	adab1_smote_sigmoid-T_{cs}^{i} -postreg	8.09	12.43	0.54	0.27	0.32	8733.03	12789.16	0.07
2018/04	0.5	1	xrf_ros-isotonic-T_{cs}^{i} -postreg	8.09	33.34	0.84	0.09	0.26	6777.96	9218.97	0.48
2018/04	rand	1	wxrf_ros_sigmoid-T_{cs}^{i} -mean	8.09	43.55	0.98	0.01	0.55	3735.29	8297.38	13.81
2018/05	1	1	adab2_rus-isotonic-T_{cs}^{i} -postreg	8.01	12.51	0.82	0.07	0.26	9765.23	13119.91	2.39

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2018/05	0.5	1	rusboost_none_none-T_{cs}^{i}-mean	8.01	36.65	0.96	0.02	0.24	7596.42	9939.64	3.77
2018/05	rand	1	bal_rf_none_sigmoid-T_{cs}^{i}-postreg	8.01	48.01	0.91	0.02	0.30	6588.44	9386.79	3.49
2018/06	1	1	adab2_ros_sigmoid-T_{cs}^{i}-postreg	9.05	13.88	0.80	0.06	0.23	13708.92	17892.66	3.52
2018/06	0.5	1	adab2_none_sigmoid-T_{cs}^{i}-postreg	9.05	37.53	0.91	0.01	0.17	11569.85	13893.63	6.73
2018/06	rand	1	adab1_none-isotonic-T_{cs}^{i}-postreg	9.05	47.53	0.92	0.02	0.67	4281.80	13173.46	6.24
2018/07	1	1	adab2_rus-isotonic-T_{cs}^{i}-mean	6.87	13.34	0.89	0.03	0.20	7750.51	9747.05	5.96
2018/07	0.5	1	rf_ros_none-T_{cs}^{i}-mean	6.87	35.89	0.95	0.01	0.18	5556.31	9747.05	13.39
2018/07	rand	3	xrf_smote-isotonic-T_{cs}^{i}-postreg	6.25	20.19	0.59	0.27	0.78	1481.94	6828.25	0.11
2018/08	1	1	xrf_none_none-T_{cs}^{i}-postreg	6.72	14.15	0.79	0.07	0.15	9918.14	11708.51	1.90
2018/08	0.5	1	wrf_none_sigmoid-T_{cs}^{i}-postreg	6.72	37.27	0.94	0.01	0.16	7306.00	8700.07	11.39
2018/08	rand	1	bal_rf_none-isotonic-T_{cs}^{i}-postreg	6.72	47.90	0.87	0.02	0.23	6174.96	8053.30	3.75
2018/09	1	1	easy_ensmb_none_none-$T = 0.5$	9.28	11.76	0.50	0.24	0.36	11172.95	17555.27	0.45
2018/09	0.5	1	bal_rf_none-isotonic-$T = 0.5$	9.28	33.78	0.83	0.05	0.48	9062.25	17555.27	2.36
2018/09	rand	1	bal_rf_none-isotonic-T_{cs}^{i}-postreg	9.28	45.52	0.86	0.04	0.57	7594.42	17555.27	2.54
2018/	1	1	adab1_rus_sigmoid-	11.5	11.27	0.4	0.3	0.2	12463.	16772.5	0.21

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10			T_{cs}^{i} -postreg	7		5	3	6	05	3	
2018/10	0.5	1	wrf_none_none-T_{cs}^{i} -postreg	11.57	32.20	0.92	0.04	0.15	9743.45	11519.75	3.46
2018/10	rand	1	rdf_smote_sigmoid-T_{cs}^{i} -postreg	11.57	44.39	0.91	0.04	0.71	2729.36	9273.33	4.47
2018/11	1	1	xrf_smote_none-T_{cs}^{i} -postreg	10.86	10.97	0.65	0.20	0.16	12671.80	15114.80	0.78
2018/11	0.5	1	easy_ensmb_none_none-T_{cs}^{i} -mean	10.86	32.09	0.95	0.03	0.17	8649.31	10381.59	2.52
2018/11	rand	1	adab2_none-isotonic-T_{cs}^{i} -postreg	10.86	43.67	0.88	0.05	0.07	8496.65	9133.13	2.35
2018/12	1	1	easy_ensmb_none_sigmoid-T_{cs}^{i} -postreg	8.95	11.37	0.72	0.12	0.17	9969.19	12006.99	1.47
2018/12	0.5	1	wrf_none-isotonic-T_{cs}^{i} -postreg	8.95	31.98	0.90	0.02	0.11	7109.03	7964.33	5.29
2018/12	rand	1	rf_none_sigmoid-T_{cs}^{i} -postreg	8.95	45.67	0.92	0.00	0.15	5932.89	6994.47	6.19
2019/01	1	1	wrf_rus_none-T_{cs}^{i} -postreg	12.37	9.36	0.65	0.24	0.14	12184.36	14212.21	0.64
2019/01	0.5	1	rdf_none_sigmoid-T_{cs}^{i} -postreg	12.37	30.17	0.94	0.02	0.04	8707.68	9084.91	9.15
2019/01	rand	1	adab1_none-isotonic-T_{cs}^{i} -postreg	12.37	42.44	0.95	0.02	0.72	2282.78	8058.50	7.78
2019/02	1	1	adab1_rus_sigmoid-T_{cs}^{i} -postreg	14.69	10.41	0.52	0.26	0.16	12839.96	15316.98	0.83
2019/02	0.5	1	bal_rf_none_sigmoid-T_{cs}^{i} -postreg	14.69	31.72	0.79	0.07	0.08	10378.46	11346.17	2.26
2019/02	rand	1	adab2_none_sigmoid-	14.69	44.57	0.84	0.06	0.19	8279.77	10169.78	2.45

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			T_{cs}^{i} -postreg								
2019/03	1	1	easy_ensmb_none-isotonic-T_{cs}^{i} -postreg	11.62	12.17	0.58	0.19	0.32	10038.56	14837.82	0.89
2019/03	0.5	3	adab1_ros_sigmoid-T_{cs}^{i} -postreg	11.62	6.69	0.36	0.39	0.25	5005.78	6637.93	0.23
2019/03	rand	1	adab2_rus-isotonic-T_{cs}^{i} -postreg	11.62	47.45	0.83	0.08	0.32	5653.64	8300.72	1.20

Figure 1: Cost performance of the cost-sensitive models across all prediction periods and interventions. The bars represent the total misclassification cost of the top-ranking model (ranked by *CIS*). The solid line shows the cost of the benchmark under the same operating conditions.

5.2. Absenteeism prediction when costs are unknown

As was mentioned in remark 2, the value of parameter $\tilde{t}_s$ is currently unknown. In order to provide a viable alternative to the practitioner, we assume equal misclassification costs and perform model selection using the balanced accuracy score (*BACC*). Table 6 presents the results of the top-performing cost-insensitive models in every prediction period. We note that there is some variation in model performance across periods, which we attribute to changes in the class distribution across different months.

Table 6: Model selection when ranking by balanced accuracy. *CIS* for each combination of intervention and $\tilde{t}_s$ are included for reference. Negative *CIS* values indicate that the model is more costly than the benchmark solution.

Target period	Method algo_sampling_calibr-T	%pos	Error-based metrics				CIS			
			FNR	FPR	BACC	AUC	$\tilde{t}_s$	Case 1	Case 2	Case 3
2018/03	adab1_none_isotonic-Tbest	15.57	0.42	0.33	0.62	0.67	1	0.22	-3.98	-0.67
							0.5	0.23	-2.56	-0.23

							rand	0.15	-2.38	-0.18
2018/04	easy_ensmb_none_none-Tbest	8.13	0.40	0.35	0.63	0.67	1	0.24	-1.67	-0.01
							0.5	0.12	-0.96	0.21
							rand	0.06	-0.77	0.23
2018/05	easy_ensmb_none_none-Tbest	8.0	0.37	0.36	0.63	0.68	1	0.00	-3.16	-0.47
							0.5	-0.19	-2.53	-0.28
							rand	-0.84	-2.33	-0.23
2018/06	easy_ensmb_none_none-Tbest	9.12	0.37	0.34	0.65	0.70	1	0.23	-3.68	-0.61
							0.5	0.19	-2.88	-0.36
							rand	0.16	-2.62	-0.35
2018/07	adab1_none_isotonic-Tbest	6.87	0.42	0.31	0.63	0.68	1	0.04	-1.87	-0.05
							0.5	-0.16	-1.29	0.13
							rand	-0.30	-1.12	0.16
2018/08	adab1_ros_none-Tbest	6.8	0.42	0.27	0.65	0.71	1	0.11	-2.73	-0.28
							0.5	-0.01	-2.07	-0.07
							rand	-0.17	-1.85	-0.04
2018/0	easy_ensmb_none_none-Tbes	9.26	0.45	0.2	0.63	0.68	1	0.34	-3.1	-0.4

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9	t			9					4	0
							0.5	0.45	-2.23	-0.12
							rand	0.49	-2.08	-0.05
2018/10	adab1_ros_isotonic-0.5	11.61	0.42	0.32	0.63	0.68	1	0.28	-3.30	-0.47
							0.5	0.14	-2.23	-0.14
							rand	0.10	-2.06	-0.09
2018/11	adab1_ros_none-0.5	10.86	0.41	0.32	0.63	0.68	1	0.15	-3.87	-0.64
							0.5	-0.02	-2.89	-0.34
							rand	-0.14	-2.73	-0.30
2018/12	adab1_none_isotonic-Tbest	9.04	0.35	0.36	0.64	0.69	1	0.10	-2.20	-0.18
							0.5	-0.15	-1.45	0.06
							rand	-0.19	-1.17	0.10
2018/01	easy_ensmb_none_sigmoid-Tbest	12.28	0.45	0.31	0.62	0.66	1	0.12	-3.81	-0.61
							0.5	-0.09	-2.56	-0.22
							rand	-0.23	-2.35	-0.13
2018/02	easy_ensmb_none_isotonic-Tbest	14.75	0.39	0.35	0.63	0.67	1	0.15	-4.54	-0.85
							0.5	0.11	-3.21	-0.44

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							rand	0.04	-2.73	-0.34
2018/03	easy_ensmb_none_none-Tbest	11.71	0.39	0.32	0.65	0.70	1	0.28	-2.18	-0.13
							0.5	0.12	-1.21	0.17
							rand	-0.081	-1.12	0.22

Figure 2: Comparison of top models' cost performance across prediction periods and interventions. The solid line represents the best performing cost-sensitive models (ranked by *CIS*). The dotted line shows the best performing cost-insensitive model (ranked by *BACC*) under the same operating conditions. The horizontal line shows $CIS = 0$.

When we compare our results to the literature, we find that a number of studies demonstrate higher performance under more severe class imbalance. For example [10] reports *AUC* 0.76 in a model of predicting sick leave due musculoskeletal disorders at the horizon of 3 months having less than 1 % positives in their sample. Their predictors included musculoskeletal complaints, burnout, distress, among others. [12] use measures of depressed mood, distress and fatigue. [11] uses self-rated health, mental health factors and psychosocial work characteristics. [13] shows the highest performance *AUC* 0.86 (10% positives) using an attitudinal predictor called the Work Ability Index. We conclude that when the misclassification costs are unknown, some objective health-related predictors may be necessary to achieve better performance.

5.2.1. Cost performance of cost-insensitive models

To investigate the cost performance of the top-ranking cost-insensitive models, we calculated cost metrics under all nine intervention scenarios. Figure 2 demonstrates the difference in cost performance achieved by the top-ranking cost-insensitive models when ranked by *BACC* versus top-ranking cost-sensitive models. Despite reasonable performance, when evaluated using error-based metrics, cost-insensitive models rarely have positive *CIS* and are always inferior to cost-sensitive models.

Model selection based on *BACC* when costs are unknown does not appear to be a viable

solution based on costs derived using our artificial $\tilde{t}_s$. Instead, effort should be put into determining $\tilde{t}_s$ to facilitate the use of misclassification costs. Of course, once this is done, this statement can be revisited.

6. Conclusion

In this paper we described a potential solution to a real-world problem, requested by HR and Well-being specialists. We were provided with an anonymised dataset containing employee payroll information only, without any health-related data. This carried the risk of the resulting models be based on correlations unrelated to health. We therefore would like to emphasise, that our conceptual model works under the assumption that any intervention considered aims to *increase employee well-being*. Thus targeting someone erroneously should never lead to a negative outcome for that individual.

Our focus was on developing a flexible, model-agnostic framework to predict short-term employee sickness absence using HR and payroll data only. Thanks to focusing on existing open-source implementations, our model selection and evaluation framework can be readily deployed to the cloud. We provided two alternative evaluation metrics under class and cost imbalance: with and without knowledge of costs. We demonstrate that improved cost-effectiveness of intervention campaigns can be achieved by focussing classifiers on the specific operating conditions of the underlying training data.

Our main take-away for practitioners is as follows:

- In the absence of misclassification costs, balanced accuracy offers more transparency over AUC, but cost-based metrics almost always achieve higher savings. Models selected based on $BACC$ are not a cost-effective alternative to those selected on cost-based metrics. Thus efforts should be directed to quantifying the effect interventions have on individuals so that misclassification costs can be specified.
- When the expected response to an intervention is known, $DMECC$ allows to assess the cost-effectiveness of an intervention for each observation resulting in a more profitable intervention campaign.
- Investigating the cost distribution on the available training data can allow one to conduct a preliminary cost and benefit analysis of a given intervention.

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6.1. Limitations and future research

The experiments presented in this paper are not without limitations. Firstly, they hinge on the *a priori* knowledge of the individual response to the intervention, which in practice can rarely be obtained. Instead of assigning arbitrary values, this parameter could be estimated, with the help of causal inference models. Causal models estimate individual-level treatment effects from observational data or randomized control trials, an example of algorithmic causal modelling can be found in [59]. Related to that, ways to improve the estimates of expected hours of absence per individual might also be worth revisiting.

Secondly, we assume that the effect of any given intervention does not extend beyond the prediction period. This can be rectified by either considering a longer prediction horizon (e.g. a quarter), or by extending the conceptual model to a dynamic setting. This, combined with an estimation of the expected duration of the individual response to a treatment might be an interesting avenue for future research.

Thirdly, we assume that employees with varying hours of sickness will have identical response to any given intervention. Without the knowledge of the true cause of absence, it is not realistic to treat, for example, an employee with 10 hours of absence and an employee with 100 hours of absence equally. An interesting direction could be instead to consider multi-class setting and to evaluate an array of interventions, finding the optimal one per individual. Lastly, repeating the experiments with objective health-related predictors would be desirable.

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Highlights

- A flexible framework for employee absenteeism prediction to support HR decisions.
- We predict employees at risk of absence without having health variables.
- We use cost-sensitive learning to optimise investment into well-being interventions.
- Conceptual model of employee sickness absence misclassification costs is developed.
- We showcase the application of our conceptual model and evaluate using cost metrics.

Author biography

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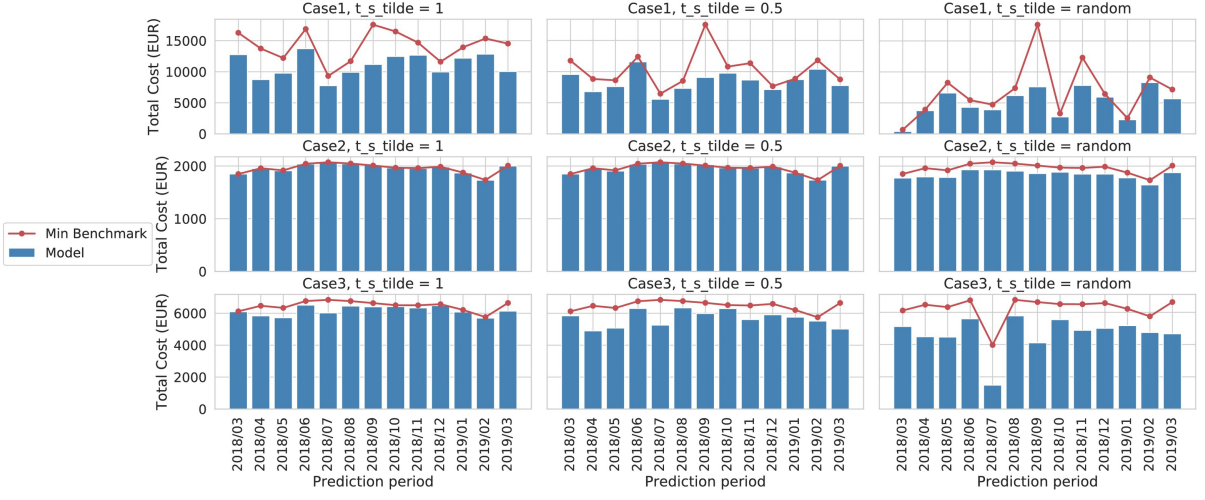

Case1, t_s_tilde = 1
Case1, t_s_tilde = 0.5
Case1, t_s_tilde = random
Case2, t_s_tilde = 1
Case2, t_s_tilde = 0.5
Case2, t_s_tilde = random
Case3, t_s_tilde = 1
Case3, t_s_tilde = 0.5
Case3, t_s_tilde = random
Total Cost (EUR)
Min Benchmark
Model
Prediction period
2018/03
2018/04
2018/05
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Figure 1

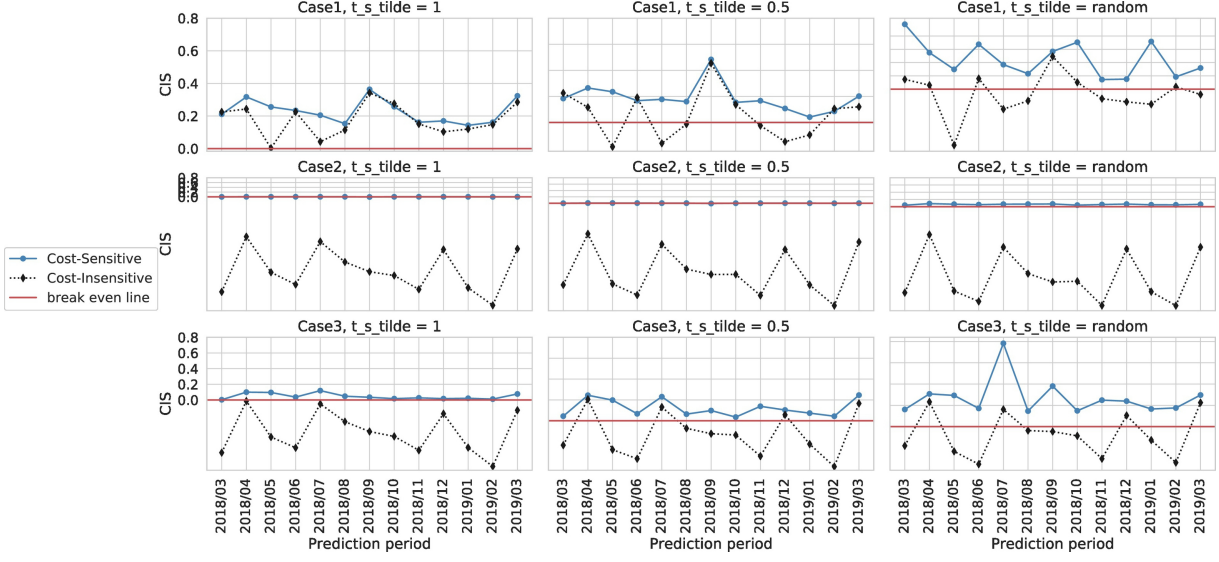

Case1, t_s_tilde = 1
Case1, t_s_tilde = 0.5
Case1, t_s_tilde = random
Case2, t_s_tilde = 1
Case2, t_s_tilde = 0.5
Case2, t_s_tilde = random
Case3, t_s_tilde = 1
Case3, t_s_tilde = 0.5
Case3, t_s_tilde = random
CIS
Prediction period
Cost-Sensitive
Cost-Insensitive
break even line
2018/03
2018/04
2018/05
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2019/03

Figure 2